In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from io import StringIO

data = """CustomerID,Gender,Age,Annual Income (k$),Spending Score (1-100)
1,Male,19,15,39
2,Male,21,15,81
3,Female,20,16,6
4,Female,23,16,77
5,Female,31,17,40
6,Female,22,17,76
7,Female,35,18,6
8,Female,23,18,94
9,Male,64,19,3
10,Female,30,19,72
11,Male,67,19,14
12,Female,35,19,99
13,Female,58,20,15
14,Female,24,20,77
15,Male,37,20,13
16,Male,22,20,79
17,Female,35,21,35
18,Male,20,21,66
19,Male,52,23,29
20,Female,35,23,98
21,Male,35,24,35
22,Male,25,24,73
23,Female,46,25,5
24,Male,31,25,73
25,Female,54,28,14
26,Male,29,28,82
27,Female,45,28,32
28,Male,35,28,61
29,Female,40,29,31
30,Female,23,29,87
31,Male,60,30,4
32,Female,21,30,73
33,Male,53,33,4
34,Male,18,33,92
35,Female,49,33,14
36,Female,21,33,81
37,Female,42,34,17
38,Female,30,34,73
39,Female,36,37,26
40,Female,20,37,75
41,Female,65,38,35
42,Male,24,38,92
43,Male,48,39,36
44,Female,31,39,61
45,Female,49,39,28
46,Female,24,39,65
47,Female,50,40,55
48,Female,27,40,47
49,Female,29,40,42
50,Female,31,40,42
51,Female,49,42,52
```

```
52,Male,33,42,60
53,Female,31,43,54
54,Male,59,43,60
55,Female,50,43,45
56,Male,47,43,41
57,Female,51,44,50
58,Male,69,44,46
59,Female,27,46,51
60,Male,53,46,46
61,Male,70,46,56
62,Male,19,46,55
63,Female,67,47,52
64,Female,54,47,59
65,Male,63,48,51
66,Male,18,48,59
67,Female,43,48,50
68,Female,68,48,48
69,Male,19,48,59
70,Female,32,48,47
71,Male,70,49,55
72,Female,47,49,42
73,Female,60,50,49
74,Female,60,50,56
75,Male,59,54,47
76,Male,26,54,54
77,Female,45,54,53
78,Male,40,54,48
79,Female,23,54,52
80,Female,49,54,42
81,Male,57,54,51
82,Male,38,54,55
83,Male,67,54,41
84,Female,46,54,44
85,Female,21,54,57
86,Male,48,54,46
87,Female,55,57,58
88,Female,22,57,55
89,Female,34,58,60
90,Female,50,58,46
91,Female,68,59,55
92,Male,18,59,41
93,Male,48,60,49
94,Female,40,60,40
95,Female,32,60,42
96,Male,24,60,52
97,Female,47,60,47
98,Female,27,60,50
99,Male,48,61,42
100,Male,20,61,49
101,Female,23,62,41
102,Female,49,62,48
103,Male,67,62,59
104,Male,26,62,55
105,Male,49,62,56
106,Female,21,62,42
107,Female,66,63,50
108,Male,54,63,46
109,Male,68,63,43
110,Male,66,63,48
111,Male,65,63,52
```

```
112,Female,19,63,54
113,Female,38,64,42
114,Male,19,64,46
115,Female,18,65,48
116,Female,19,65,50
117,Female,63,65,43
118,Female,49,65,59
119,Female,51,67,43
120,Female,50,67,57
121,Male,27,67,56
122,Female,38,67,40
123,Female,40,69,58
124,Male,39,69,91
125,Female,23,70,29
126,Female,31,70,77
127,Male,43,71,35
128,Male,40,71,95
129,Male,59,71,11
130,Male,38,71,75
131,Male,47,71,9
132,Male,39,71,75
133,Female,25,72,34
134,Female,31,72,71
135,Male,20,73,5
136,Female,29,73,88
137,Female,44,73,7
138,Male,32,73,73
139,Male,19,74,10
140,Female,35,74,72
141,Female,57,75,5
142,Male,32,75,93
143,Female,28,76,40
144,Female,32,76,87
145,Male,25,77,12
146,Male,28,77,97
147,Male,48,77,36
148,Female,32,77,74
149,Female,34,78,22
150,Male,34,78,90
151,Male,43,78,17
152,Male,39,78,88
153,Female,44,78,20
154,Female,38,78,76
155,Female,47,78,16
156,Female,27,78,89
157,Male,37,78,1
158,Female,30,78,78
159,Male,34,78,1
160,Female,30,78,73
161,Female,56,79,35
162,Female,29,79,83
163,Male,19,81,5
164,Female,31,81,93
165,Male,50,85,26
166,Female,36,85,75
167,Male,42,86,20
168,Female,33,86,95
169,Female,36,87,27
170,Male,32,87,63
171,Male,40,87,13
```

```
172,Male,28,87,75
173,Male,36,87,10
174,Male,36,87,92
175,Female,52,88,13
176,Female,30,88,86
177,Male,58,88,15
178,Male,27,88,69
179,Male,59,93,14
180,Male,35,93,90
181,Female,37,97,32
182,Female,32,97,86
183,Male,46,98,15
184,Female,29,98,88
185,Female,41,99,39
186,Male,30,99,97
187,Female,54,101,24
188,Male,28,101,68
189,Female,41,103,17
190,Female,36,103,85
191,Female,34,103,23
192,Female,32,103,69
193,Male,33,113,8
194,Female,38,113,91
195,Female,47,120,16
196,Female,35,120,79
197,Female,45,126,28
198,Male,32,126,74
199,Male,32,137,18
200,Male,30,137,83"""

df = pd.read_csv(StringIO(data))

df
```

Out[2]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [3]:
```
df.info()
```

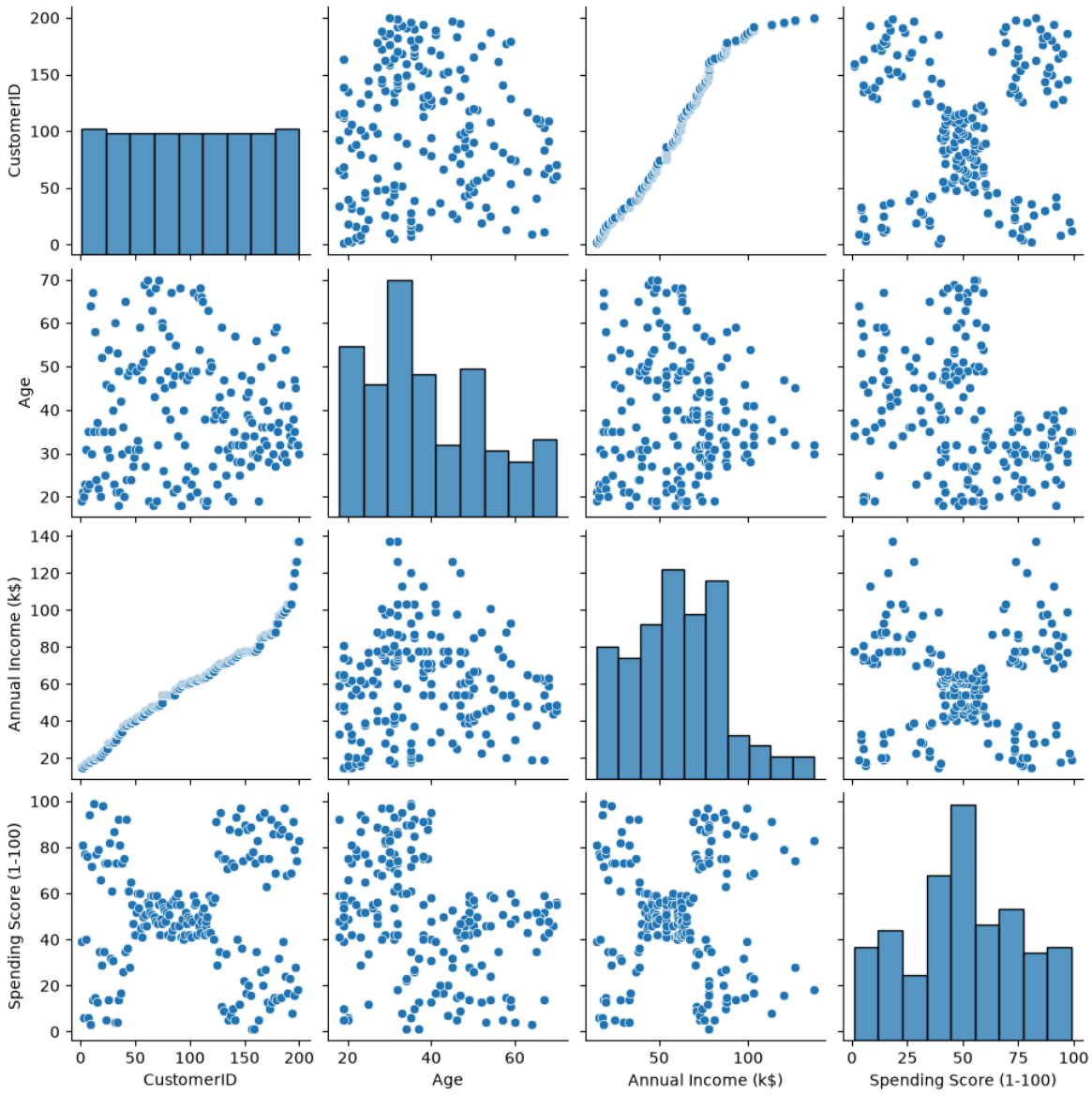
```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    str
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), str(1)
memory usage: 8.9 KB
```

In [4]:
```
df.head()
```

Out[4]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [5]:
```
sns.pairplot(df)
plt.show()
```



In [6]:
```
features = df.iloc[:, [3, 4]].values

features
```

Out[6]:
```
array([[ 15,  39],
       [ 15,  81],
       [ 16,   6],
       [ 16,  77],
       [ 17,  40],
       [ 17,  76],
       [ 18,   6],
       [ 18,  94],
       [ 19,   3],
       [ 19,  72],
       [ 19,  14],
       [ 19,  99],
       [ 20,  15],
       [ 20,  77],
       [ 20,  13],
       [ 20,  79],
       [ 21,  35],
       [ 21,  66],
       [ 23,  29],
       [ 23,  98],
       [ 24,  35],
       [ 24,  73],
       [ 25,   5],
       [ 25,  73],
       [ 28,  14],
       [ 28,  82],
       [ 28,  32],
       [ 28,  61],
       [ 29,  31],
       [ 29,  87],
       [ 30,   4],
       [ 30,  73],
       [ 33,   4],
       [ 33,  92],
       [ 33,  14],
       [ 33,  81],
       [ 34,  17],
       [ 34,  73],
       [ 37,  26],
       [ 37,  75],
       [ 38,  35],
       [ 38,  92],
       [ 39,  36],
       [ 39,  61],
       [ 39,  28],
       [ 39,  65],
       [ 40,  55],
       [ 40,  47],
       [ 40,  42],
       [ 40,  42],
       [ 42,  52],
       [ 42,  60],
       [ 43,  54],
       [ 43,  60],
       [ 43,  45],
       [ 43,  41],
       [ 44,  50],
       [ 44,  46],
       [ 46,  51],
       [ 46,  46],
```

```
       [ 46,  56],
       [ 46,  55],
       [ 47,  52],
       [ 47,  59],
       [ 48,  51],
       [ 48,  59],
       [ 48,  50],
       [ 48,  48],
       [ 48,  59],
       [ 48,  47],
       [ 49,  55],
       [ 49,  42],
       [ 50,  49],
       [ 50,  56],
       [ 54,  47],
       [ 54,  54],
       [ 54,  53],
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       [ 54,  52],
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       [ 54,  55],
       [ 54,  41],
       [ 54,  44],
       [ 54,  57],
       [ 54,  46],
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       [ 57,  55],
       [ 58,  60],
       [ 58,  46],
       [ 59,  55],
       [ 59,  41],
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       [ 60,  40],
       [ 60,  42],
       [ 60,  52],
       [ 60,  47],
       [ 60,  50],
       [ 61,  42],
       [ 61,  49],
       [ 62,  41],
       [ 62,  48],
       [ 62,  59],
       [ 62,  55],
       [ 62,  56],
       [ 62,  42],
       [ 63,  50],
       [ 63,  46],
       [ 63,  43],
       [ 63,  48],
       [ 63,  52],
       [ 63,  54],
       [ 64,  42],
       [ 64,  46],
       [ 65,  48],
       [ 65,  50],
       [ 65,  43],
       [ 65,  59],
       [ 67,  43],
       [ 67,  57],
```

```
       [ 67,  56],
       [ 67,  40],
       [ 69,  58],
       [ 69,  91],
       [ 70,  29],
       [ 70,  77],
       [ 71,  35],
       [ 71,  95],
       [ 71,  11],
       [ 71,  75],
       [ 71,   9],
       [ 71,  75],
       [ 72,  34],
       [ 72,  71],
       [ 73,   5],
       [ 73,  88],
       [ 73,   7],
       [ 73,  73],
       [ 74,  10],
       [ 74,  72],
       [ 75,   5],
       [ 75,  93],
       [ 76,  40],
       [ 76,  87],
       [ 77,  12],
       [ 77,  97],
       [ 77,  36],
       [ 77,  74],
       [ 78,  22],
       [ 78,  90],
       [ 78,  17],
       [ 78,  88],
       [ 78,  20],
       [ 78,  76],
       [ 78,  16],
       [ 78,  89],
       [ 78,   1],
       [ 78,  78],
       [ 78,   1],
       [ 78,  73],
       [ 79,  35],
       [ 79,  83],
       [ 81,   5],
       [ 81,  93],
       [ 85,  26],
       [ 85,  75],
       [ 86,  20],
       [ 86,  95],
       [ 87,  27],
       [ 87,  63],
       [ 87,  13],
       [ 87,  75],
       [ 87,  10],
       [ 87,  92],
       [ 88,  13],
       [ 88,  86],
       [ 88,  15],
       [ 88,  69],
       [ 93,  14],
       [ 93,  90],
```

```
       [ 97,  32],
       [ 97,  86],
       [ 98,  15],
       [ 98,  88],
       [ 99,  39],
       [ 99,  97],
       [101,  24],
       [101,  68],
       [103,  17],
       [103,  85],
       [103,  23],
       [103,  69],
       [113,   8],
       [113,  91],
       [120,  16],
       [120,  79],
       [126,  28],
       [126,  74],
       [137,  18],
       [137,  83]])
```

In [7]:
```
from sklearn.cluster import KMeans

model = KMeans(n_clusters=5, random_state=42, n_init=10)

model.fit(features)
```

```
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
arning: KMeans is known to have a memory leak on Windows with MKL, when there are
less chunks than available threads. You can avoid it by setting the environment v
ariable OMP_NUM_THREADS=1.
  warnings.warn(
```

Out[7]:

▾ KMeans ⓘ ?

▸ Parameters

▸ Fitted attributes

▸ 5 features

In [8]:
```
Final = df.iloc[:, [3, 4]].copy()

Final["label"] = model.predict(features)

Final.head()
```

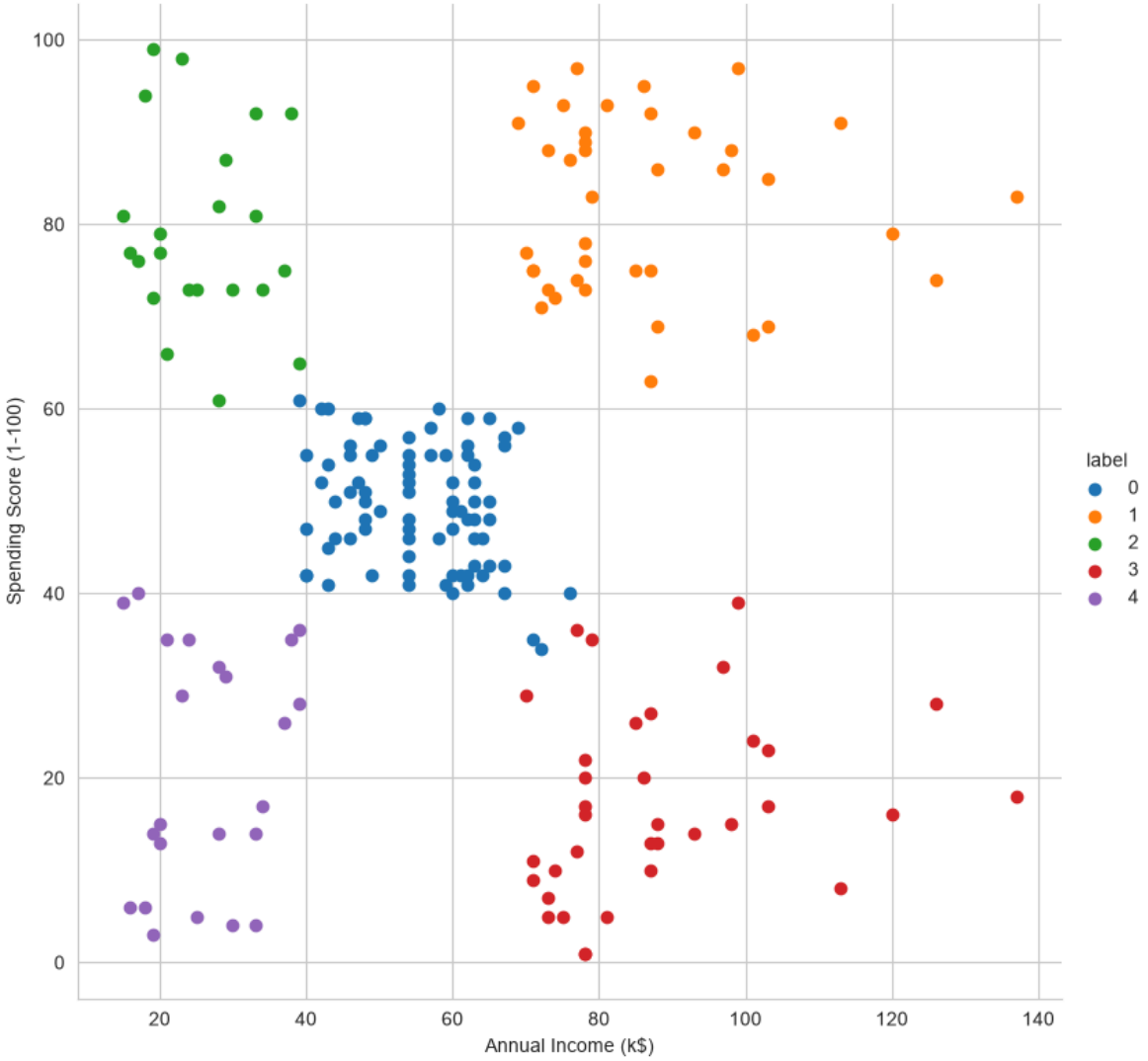
Out[8]:

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

In [9]:
```
sns.set_style("whitegrid")

sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()

plt.show()
```



In [10]:
```
features_el = df.iloc[:, [2, 3, 4]].values

wcss = []

for i in range(1, 10):
    model = KMeans(n_clusters=i, random_state=42, n_init=10)
```

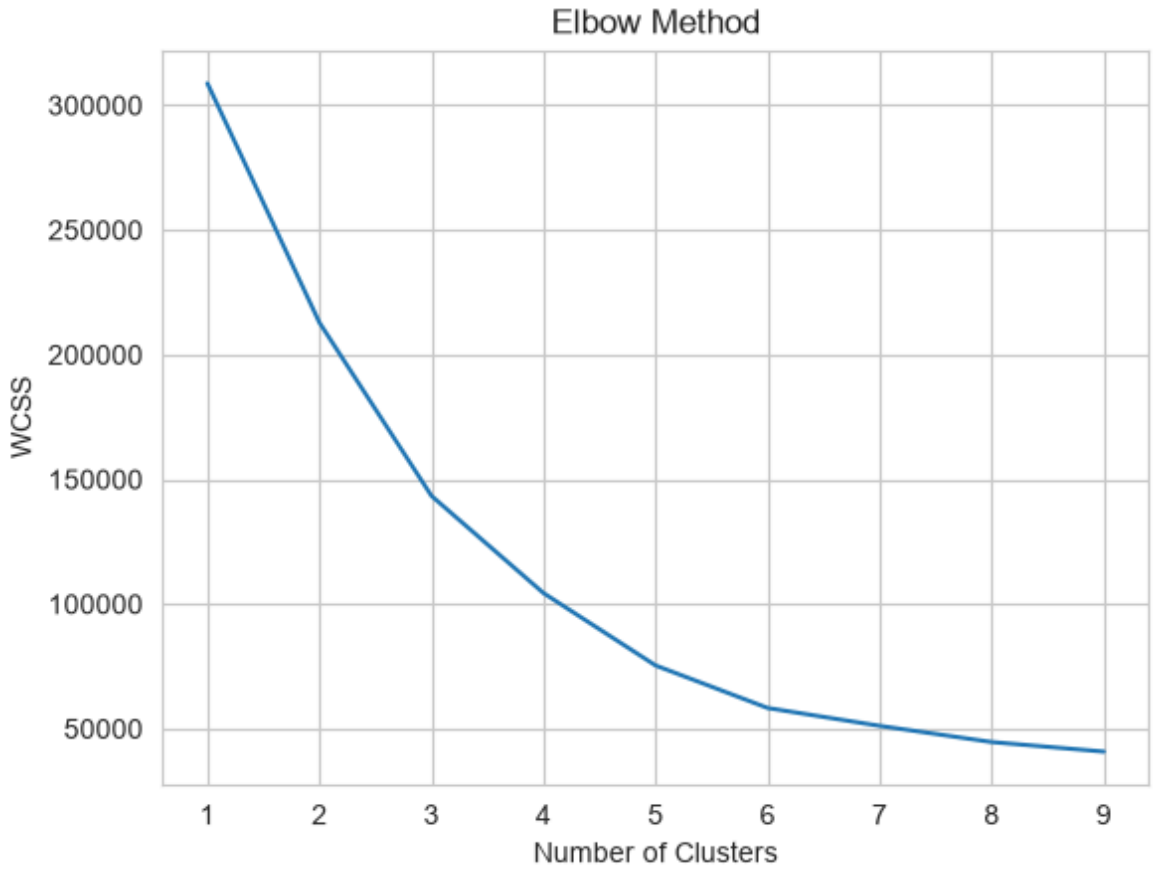
```
    model.fit(features_el)
    wcss.append(model.inertia_)
```

```
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
arning: KMeans is known to have a memory leak on Windows with MKL, when there are
less chunks than available threads. You can avoid it by setting the environment v
ariable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
arning: KMeans is known to have a memory leak on Windows with MKL, when there are
less chunks than available threads. You can avoid it by setting the environment v
ariable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
arning: KMeans is known to have a memory leak on Windows with MKL, when there are
less chunks than available threads. You can avoid it by setting the environment v
ariable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
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less chunks than available threads. You can avoid it by setting the environment v
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  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
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  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
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  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
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  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
arning: KMeans is known to have a memory leak on Windows with MKL, when there are
less chunks than available threads. You can avoid it by setting the environment v
ariable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\sasmi\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserW
arning: KMeans is known to have a memory leak on Windows with MKL, when there are
less chunks than available threads. You can avoid it by setting the environment v
ariable OMP_NUM_THREADS=1.
  warnings.warn(
```

In [11]:
```
plt.plot(range(1, 10), wcss)

plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")

plt.show()
```


Elbow Method
WCSS
300000
250000
200000
150000
100000
50000
1
2
3
4
5
6
7
8
9
Number of Clusters

In [ ]: